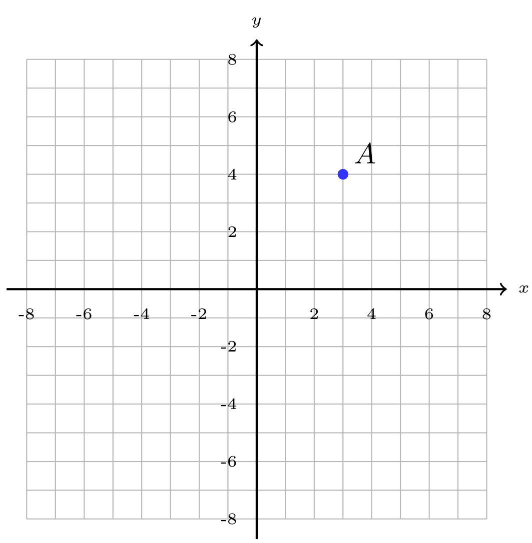
Compositions of Rigid Motions and Symmetry

Carrying out a two-step composition of rigid motions, naming the single transformation that does the same job, and describing the line and rotational symmetry of a figure

How to use this sheet.

- A *composition* is one transformation followed by another. Order matters: do the first one to the original figure, then do the second one to *that* image.
- Translations, reflections and rotations are *rigid motions*: every length, every angle and every area survives them untouched. That is what makes the checks in problems 2, 3 and 6 worth doing.
- A composition of two rigid motions is itself a single rigid motion, so it always has a one-step name. Two reflections in parallel mirrors give a translation; two reflections in intersecting mirrors give a rotation about the point where they cross.
- A figure has a *symmetry* when some rigid motion maps it exactly onto itself — a reflection in a line of symmetry, or a rotation about its centre.

1. **One point, two steps.** Point A is marked on the grid. Reflect it in the x-axis, then translate the image by the vector $\langle 2, 5 \rangle$.



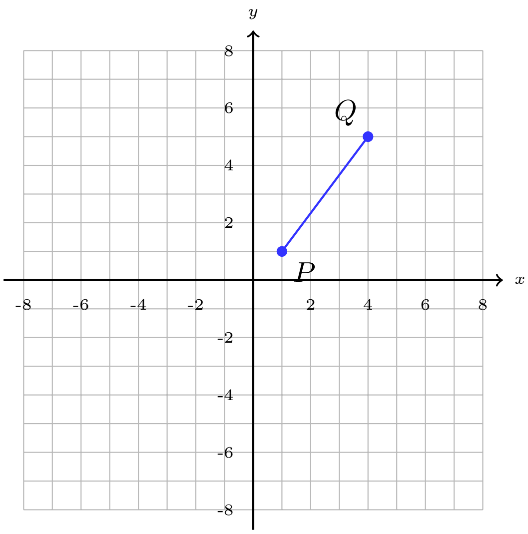
Do the two steps in the order given, one at a time, and plot both images.

After the reflection, $A' =$ (________, ________)

After the translation, $A'' =$ (________, ________)

Answer: ________________

2. A segment through a composition. Translate $\overline{PQ}$ by $\langle 3, -2\rangle$, then reflect the image in the y-axis.



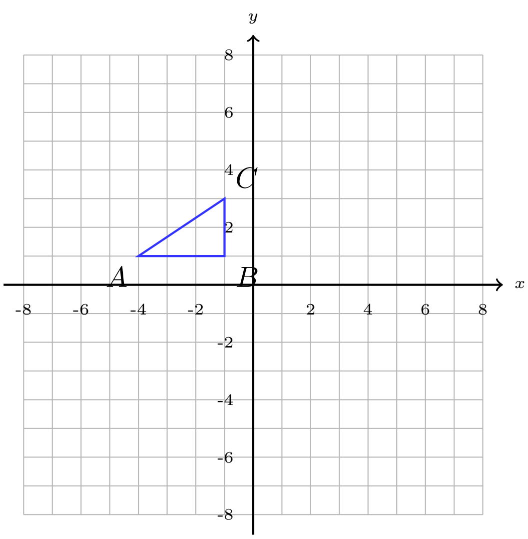
$P = (1, 1)$ and $Q = (4, 5)$.

$P'' = (\ \_\_\_\_\_\_\_\_,\ \_\_\_\_\_\_\_\_)$ $Q'' = (\ \_\_\_\_\_\_\_\_,\ \_\_\_\_\_\_\_\_)$

Now use the distance formula twice to check that the composition changed nothing about the length: find PQ and $P''Q''$.

Answer: ________________

3. A triangle through a composition. Reflect $\triangle ABC$ in the y-axis, then translate the image by $\langle 0, -4\rangle$.



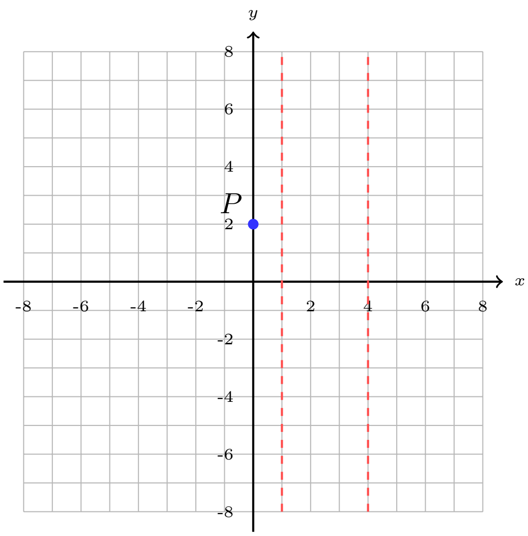
$A = (-4, 1)$, $B = (-1, 1)$, $C = (-1, 3)$. Draw the final image and label it $A''B''C''$.

(a) Length of $\overline{A''B''}$: ________

(b) Midpoint of $\overline{A''C''}$: $(\ \_\_\_\_\_\_\_\_,\ \_\_\_\_\_\_\_\_)$

Answer: ________________

4. Two parallel mirrors. Reflect P in the dashed line $x = 1$, then reflect that image in the dashed line $x = 4$.



(a) $P' = (\ \_\_\_\_\_\_\_\_,\ \_\_\_\_\_\_\_\_)$ $P'' = (\ \_\_\_\_\_\_\_\_,\ \_\_\_\_\_\_\_\_)$

(b) How far did P travel, from P to P''? ________

(c) Describe the ONE transformation that would take P to P'' in a single step. Say what kind it is, how far, and in which direction.

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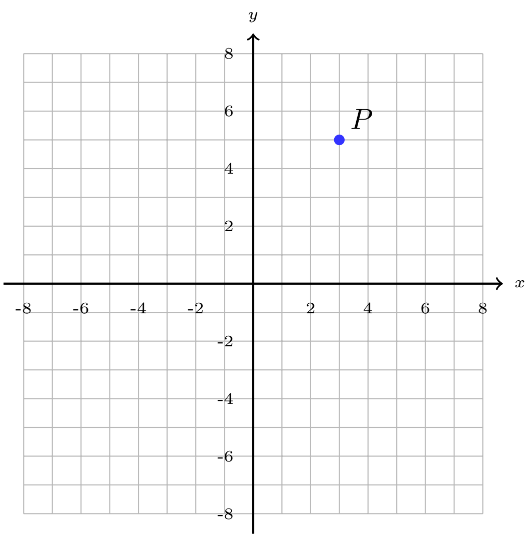
5. Two perpendicular mirrors. Reflect P in the x-axis, then reflect that image in the y-axis.



(a) $P'' = (\ \_\_\_\_\_\_\_\_,\ \_\_\_\_\_\_\_\_)$

(b) Find the midpoint of $\overline{PP''}$: $(\ \_\_\_\_\_\_\_\_,\ \_\_\_\_\_\_\_\_)$

(c) Use part (b) to describe the single transformation that does the same job. Name the kind, the angle and the centre.

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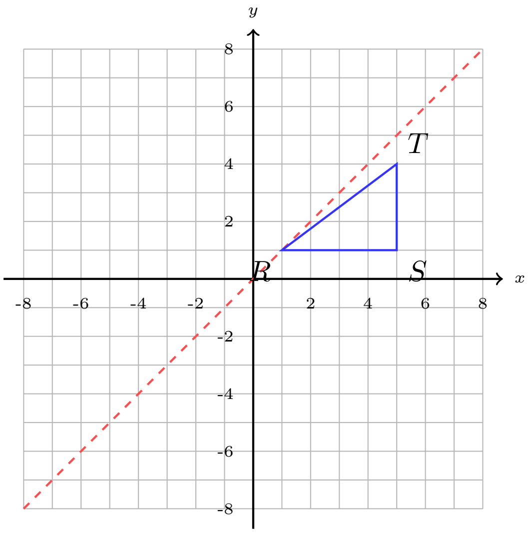
6. Mirrors that meet at 45°. Reflect $\triangle RST$ in the dashed line $y = x$, then reflect the image in the x-axis.



$R = (1, 1)$, $S = (5, 1)$, $T = (5, 4)$. Draw and label the final image.

(a) Area of $\triangle RST$: ________

(b) Area of the final image: ________

(c) Name the single transformation: kind, angle, direction and centre.

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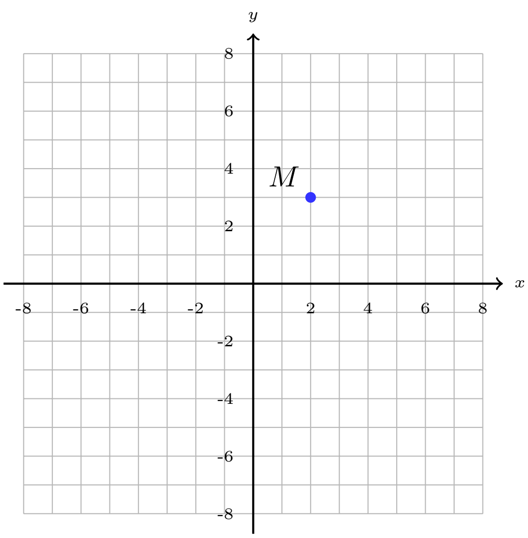
7. Translation, then a half turn. Translate M by $\langle 5, -1 \rangle$, then rotate the image 180° about the origin.



$M = (2, 3)$. Plot both images.

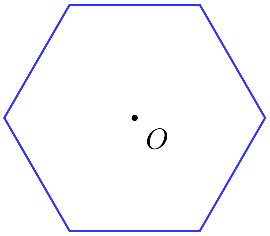
After the translation, $M' = ($ ________, ________$)$

After the half turn, $M'' = ($ ________, ________$)$

Would you get the same final point doing the two steps the other way round? ________

Answer: ________________

8. Symmetry of a regular hexagon. The hexagon below is regular: all six sides equal, all six angles equal, with centre O.



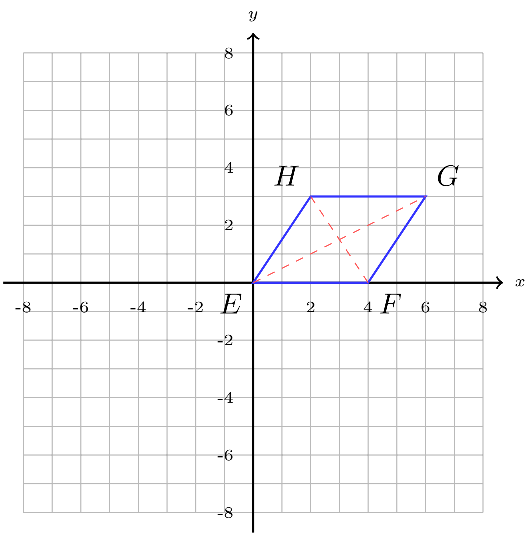
(a) Draw every line of symmetry on the figure. How many are there? _________

Some run vertex to vertex; the rest run edge-midpoint to edge-midpoint. Count each family, then double.

(b) What is the SMALLEST angle of rotation about O that maps the hexagon exactly onto itself?

Answer: _________________ degrees

9. Symmetry of a parallelogram. $EFGH$ has vertices $E(0,0)$, $F(4,0)$, $G(6,3)$ and $H(2,3)$. Its diagonals are drawn dashed.



(a) Midpoint of $\overline{EG}$: (_________, _________)

(b) Midpoint of $\overline{FH}$: (_________, _________)

(c) Explain why $EFGH$ maps onto itself under a half turn, and why it has no line of symmetry at all. Use your two midpoints.

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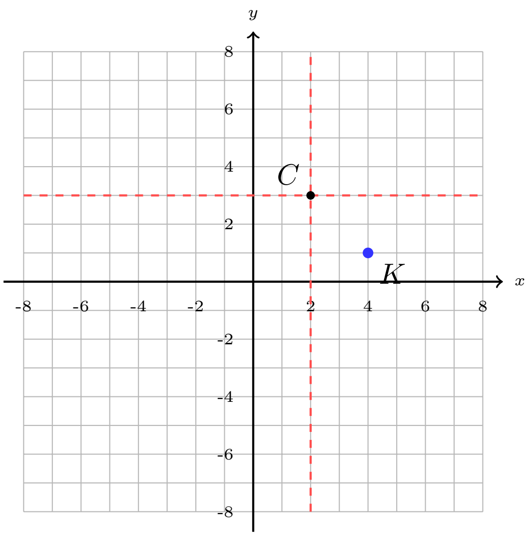
10. Why perpendicular mirrors make a half turn. The dashed lines are $x = 2$ and $y = 3$, and they meet at $(2, 3)$.



(a) Reflect $K(4, 1)$ in $x = 2$, then reflect that image in $y = 3$.

$K'' =$ (__________, __________)

(b) Midpoint of $\overline{KK''}$: (__________, __________)

(c) Explain why reflecting in ANY two perpendicular lines is always the same as a half turn about the point where they cross — not just for this K.

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